PALMYRA ATOLL · PACIFIC OCEAN

The forest below

What does it take for a forest to return?

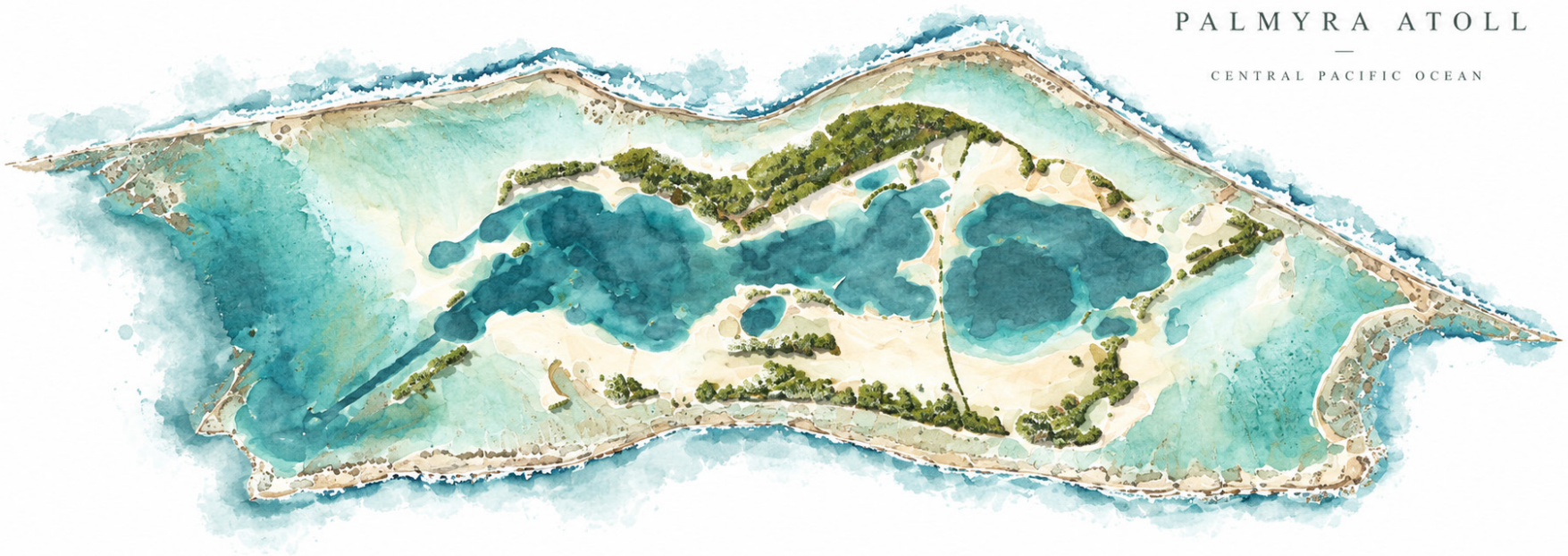
Trees, fungi, and the hidden side of forest restoration.



A VISUAL SCIENCE STORY

Where this story begins

Palmyra Atoll is a remote Pacific island ecosystem where scientists are studying how native forests return.



Here, the recovery of *Pisonia grandis* forests may depend on more than seeds and sunlight. Beneath the soil, hidden fungal relationships could be part of the story.

An island transformed

Over time, coconut palms replaced much of Palmyra's native *Pisonia grandis* forest.

But not all trees support the same life.

As native forests declined, seabirds lost important nesting habitats, disrupting the natural connections between the forest and the surrounding reefs.



Native *Pisonia* Forest
A natural habitat that supports seabird nesting and the island's ecological balance.

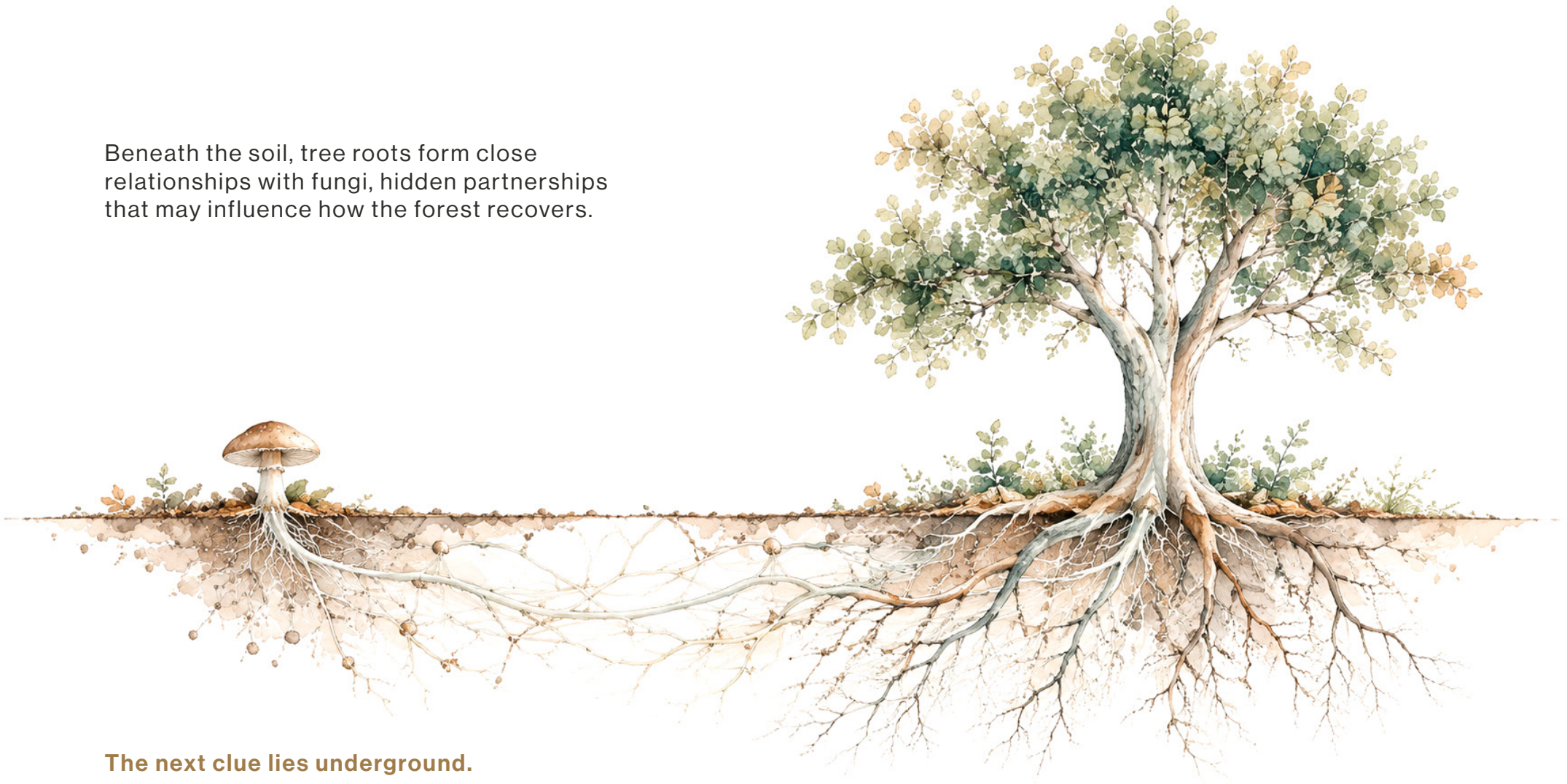


Coconut-Dominated Forest
An altered habitat offering fewer suitable nesting sites for many seabirds.

A forest is more than its trees

Removing coconut palms is only part of restoring Palmyra's native forest.

Beneath the soil, tree roots form close relationships with fungi, hidden partnerships that may influence how the forest recovers.



The next clue lies underground.

An unexpected partnership

Scientists discovered a remarkable relationship between *Pisonia grandis* and a group of fungi called *Tomentella*.

100%

of the sampled Pisonia roots were associated with these fungi, across 52 trees examined.



A hidden partnership that could be important for the forest's recovery.

A remarkable exchange

Beneath the forest, trees and fungi exchange resources essential to their survival.

Fungi help trees absorb water and nutrients from the soil. In return, trees provide the fungi with sugars produced through photosynthesis.



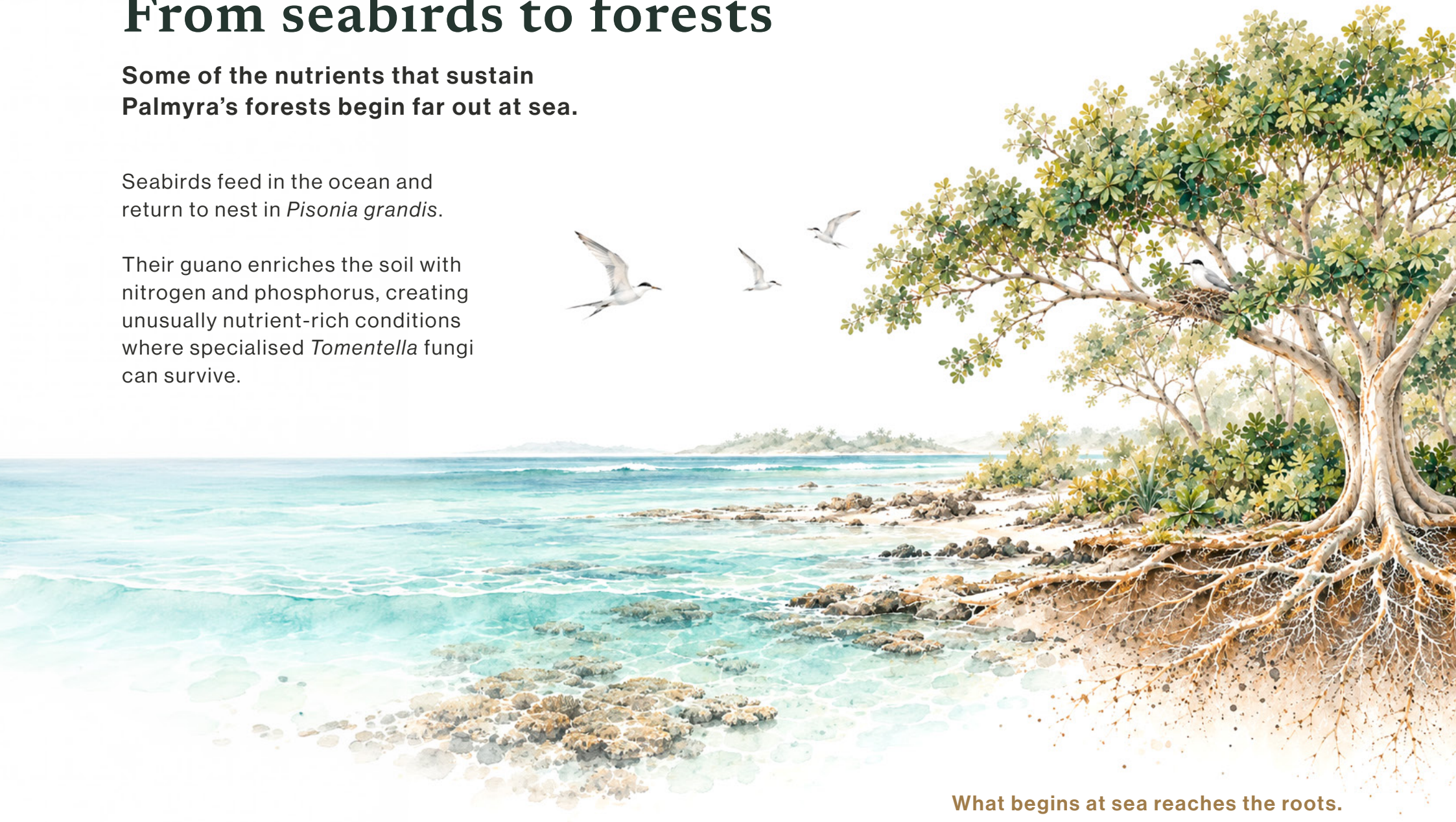
On Palmyra, this partnership thrives in extraordinary conditions.

From seabirds to forests

Some of the nutrients that sustain Palmyra's forests begin far out at sea.

Seabirds feed in the ocean and return to nest in *Pisonia grandis*.

Their guano enriches the soil with nitrogen and phosphorus, creating unusually nutrient-rich conditions where specialised *Tomentella* fungi can survive.



What begins at sea reaches the roots.

The forest feeds the sea

The influence of Palmyra's native forest extends far beyond the shoreline.

Rain carries seabird-derived nutrients from the forest into surrounding waters, supporting plankton growth and attracting manta rays to feed near native forest coastlines.



Life on land shapes life beneath the waves.

The unexpected gardeners

Palmyra's land crabs may play a surprising role in the forest's hidden ecosystem.

Researchers found significantly higher fungal diversity in soils collected from crab burrows than in surrounding soils.

By digging and mixing the ground, these animals may help create conditions that support a richer fungal community.



Even a crab's burrow can shelter an unseen world.

Restoring the unseen

Bringing back Palmyra's native forests may require restoring their hidden fungal partners, too.

Researchers found that *Tomentella* fungi become less abundant farther from existing *Pisonia grandis* trees.

This raises an important possibility: introducing native fungi alongside young trees could help forests recover in areas where these partnerships are scarce.



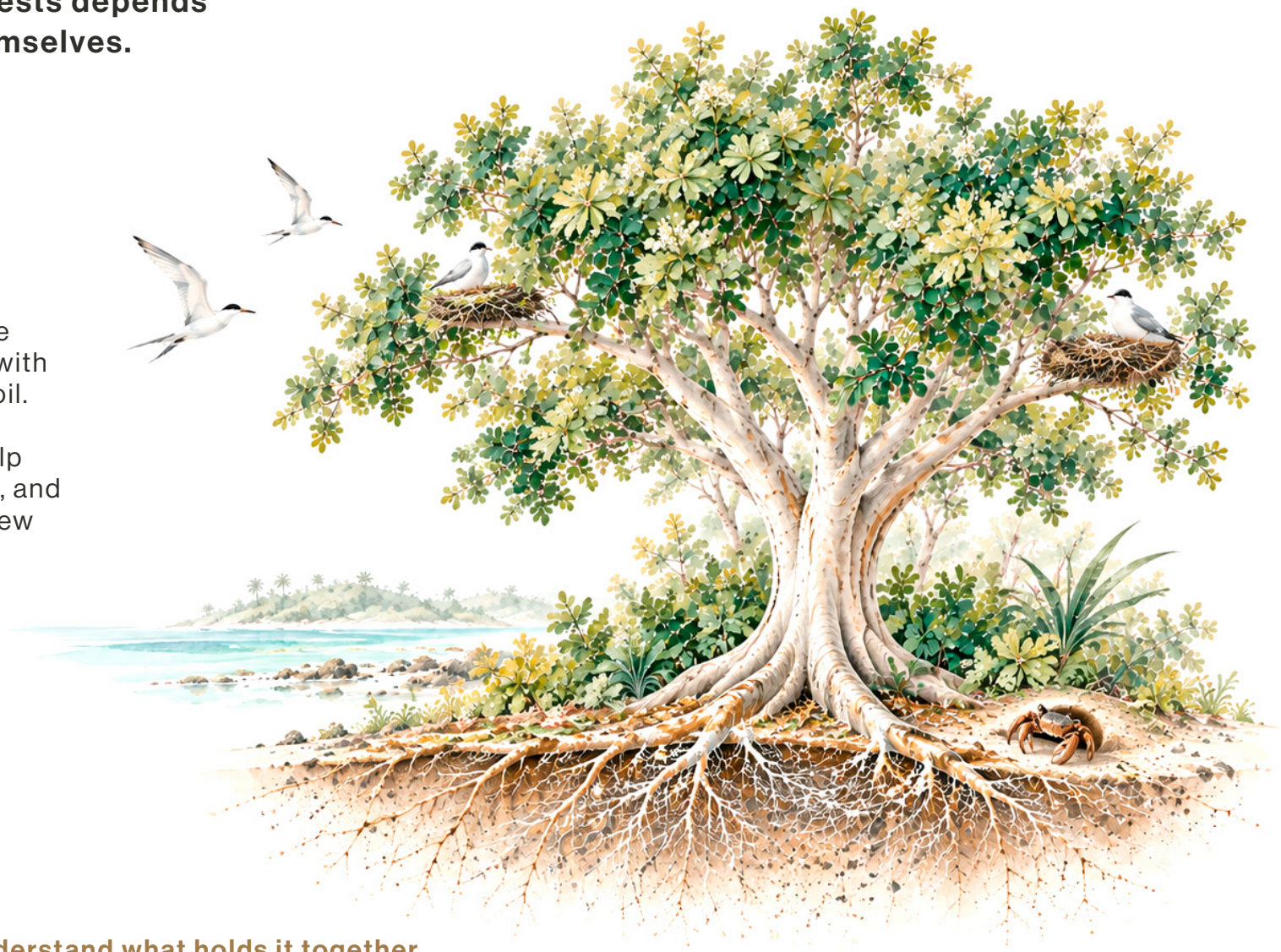
The next step is to put this idea to the test.

What holds a forest together

The future of Palmyra's forests depends on more than the trees themselves.

Seabirds bring nutrients from the ocean. Fungi form partnerships with roots. Land crabs reshape the soil.

Together, these relationships help sustain a remarkable ecosystem, and understanding them may open new possibilities for restoring fragile island forests.



To restore a forest, we must understand what holds it together.

Sources & References

SCIENTIFIC RESEARCH

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Pre-restoration woody species crown and vegetation community mapping…

CONSERVATION & BACKGROUND

The Nature Conservancy
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U.S. Fish & Wildlife Service
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Society for the Protection of Underground Networks (SPUN) (2026)
Data from Earth's most remote atoll show soil fungi are key to island regeneration.

ABOUT THIS PROJECT

An independent visual storytelling project created for educational purposes. Not commissioned by or affiliated with the organisations or researchers referenced.

Illustrations are AI-assisted artistic interpretations. Ecological processes and geographical features are not necessarily represented to scale.

A Personal Reflection

This visual story was inspired by the beauty of the research behind it.

Pisonia grandis trees, *Tomentella* fungi, seabirds, manta rays, and crabs coexist in a web of interconnected relationships on Palmyra Atoll. They are all part of the same ecosystem, as if they were one single, indivisible being.

As humans, we are part of it too. We are inseparable from it. When we disrupt the harmony of this system, we only distance ourselves from what we truly are, creating chaos and suffering.

To understand this system is to understand life itself, and the natural connections we so often insist on breaking.

I'm deeply grateful to the thousands of people who dedicate their lives to restoring the natural balance, protecting the life, beauty, and truth of this world.



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